



Analyzing the build





Analyzing the build: available tools

- ▶ Buildroot provides several useful tools to analyze the build:
 - The **licensing report**, covered in a previous section, which allows to analyze the list of packages and their licenses.
 - The **dependency graphing** tools
 - The **build time graphing** tools
 - The **filesystem size** tools

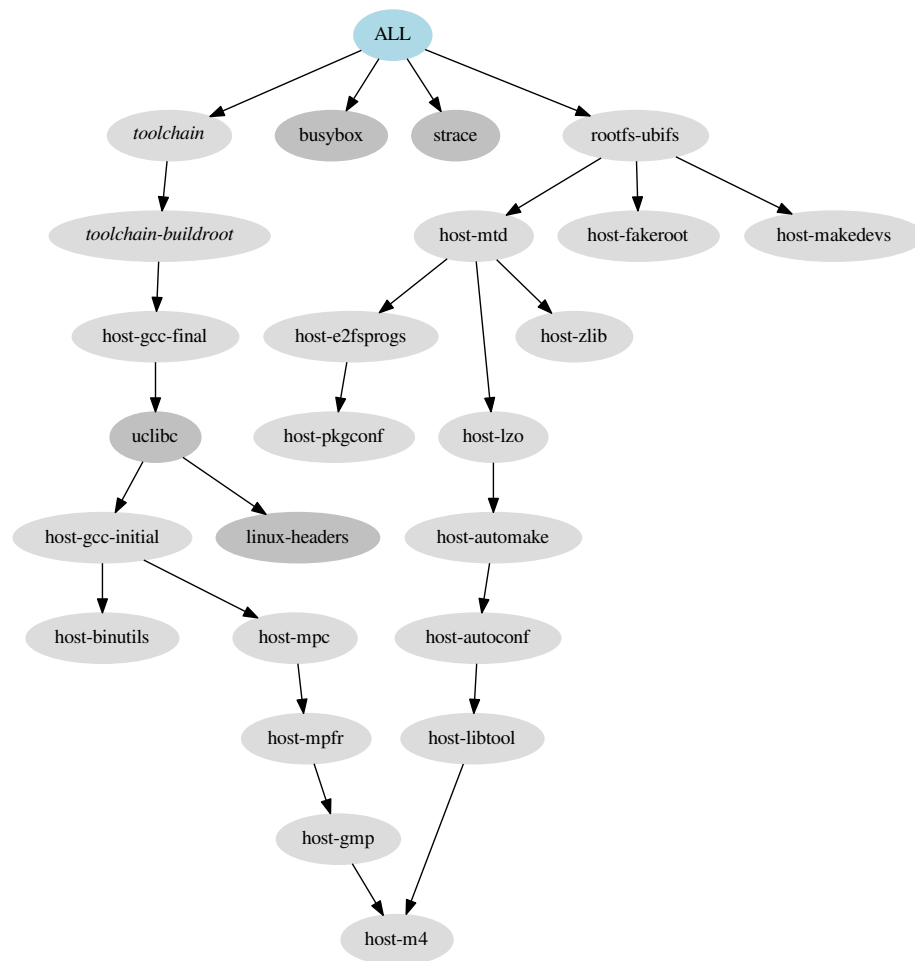


Dependency graphing

- ▶ Exploring the dependencies between packages is useful to understand
 - why a particular package is being brought into the build
 - if the build size and duration can be reduced
- ▶ `make graph-depends` to generate a full dependency graph, which can be huge!
- ▶ `make <pkg>-graph-depends` to generate the dependency graph of a given package
- ▶ The graph is done according to the current Buildroot configuration.
- ▶ Resulting graphs in `$(0)/graphs/`



Dependency graph example



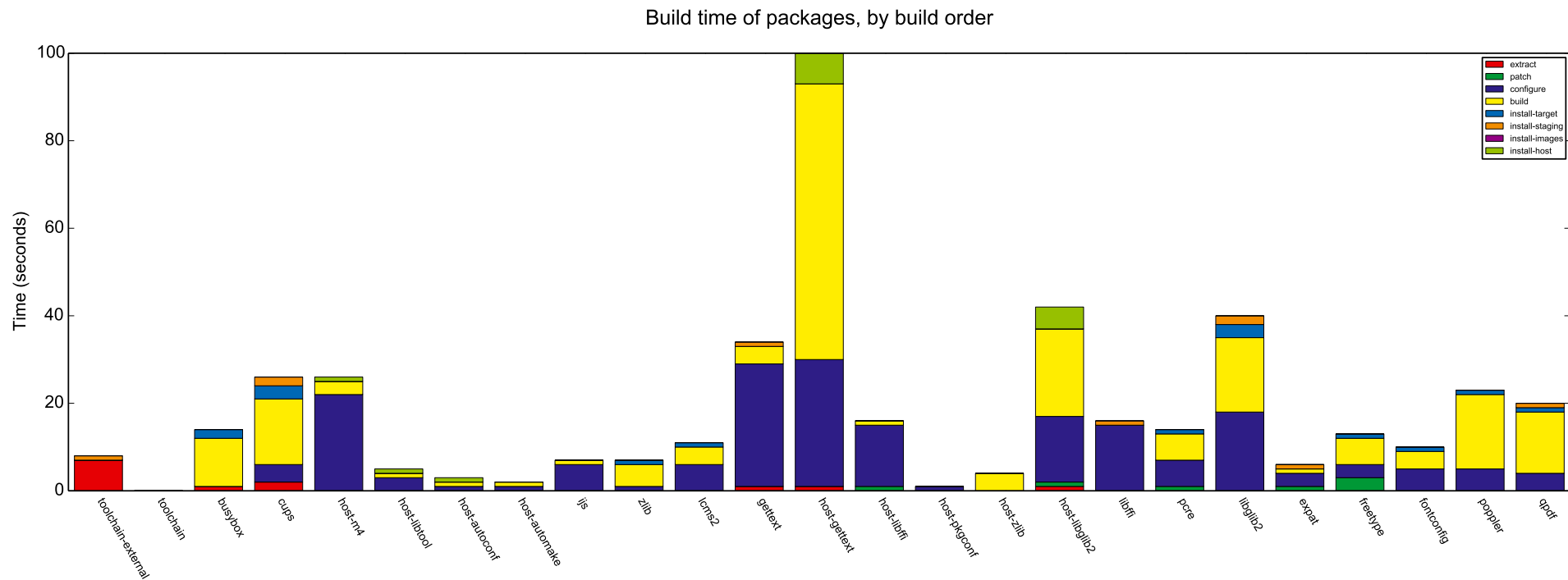


Build time graphing

- ▶ When the generated embedded Linux system grows bigger and bigger, the build time also increases.
- ▶ It is sometimes useful to analyze this build time, and see if certain packages are particularly problematic.
- ▶ Buildroot collects build duration data in the file `$(0)/build/build-time.log`
- ▶ `make graph-build` generates several graphs in `$(0)/graphs/`:
 - `build.hist-build.pdf`, build time in build order
 - `build.hist-duration.pdf`, build time by duration
 - `build.hist-name.pdf`, build time by package name
 - `build.pie-packages.pdf`, pie chart of the per-package build time
 - `build.pie-steps.pdf`, pie chart of the per-step build time
- ▶ Note: only works properly after a complete clean rebuild.



Build time graphing: example





Filesystem size graphing

- ▶ In many embedded systems, storage resources are limited.
- ▶ For this reason, it is useful to be able to analyze the size of your root filesystem, and see which packages are consuming the biggest amount of space.
- ▶ Allows to focus the size optimizations on the relevant packages.
- ▶ Buildroot collects data about the size installed by each package.
- ▶ `make graph-size` produces:
 - `file-size-stats.csv`, CSV with the raw data of the per-file size
 - `package-size-stats.csv`, CSV with the raw data of the per-package size
 - `graph-size.pdf`, pie chart of the per-package size consumption



Filesystem size graphing: example

Filesystem size per package

Total filesystem size: 3156 kB

